MEG: Generating Molecular Counterfactual explanations for toxicity analysis

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Dataset

MoleculeNet: Wu, Z. et al. ”Moleculenet: A benchmark for molecular machine learning” (2017), and in particular:

- **TOX 21 -** Qualitative toxicity measurements on 12 biological targets, including nuclear receptors and stress response pathways.
- **ToxCast -** Toxicology data for a large library of compounds based on in vitro high-throughput screening, including experiments on over 600 tasks
- **ESOL** - Water solubility data(log solubility in mols per litre) for common organic small molecules.
- **ClinTox -** Qualitative data of drugs approved by the FDA and those that have failed clinical trials for toxicity reasons.

Description

This work addresses the Explainability of AI predictions within the domain of Chemistry. This field has seen significant integration with deep learning, particularly for healthcare-related tasks such as drug discovery, toxicology, and general molecule-property prediction. A major challenge in drug design is understanding how specific structural changes—such as adding bonds or atoms—can introduce toxicity or affect the solubility of a chemical compound.

Our approach utilizes a DGN (Deep Graph Network) acting as a property predictor. The DGN is trained to map individual molecular structures to their properties (e.g., toxicity) by learning vector representations of the input graphs. Specifically, the DGN analyzes the topologies and chemical structures to output a classification score (e.g., a probability distribution for toxic vs. non-toxic classes). Building upon the DGN's predictions, we feed a target molecule into the Generator agent. The agent's goal is to modify the molecule's topology and chemical properties, strictly adhering to chemical constraints (such as valency rules), to generate a valid *molecular counterfactual.* The output of the RL agent is a new molecule designed specifically to reverse the DGN's original prediction (e.g., transforming a non-toxic input into a toxic one, or vice-versa).

The core of the work is the implementation of a reinforcement learning agent to create new ”counterfactual” molecules. Specifically, the agent sequentially modifies the original molecule by selecting actions from a discrete space, which includes three main possibilities:

1. Adding an atom;
2. Adding or modifying a bond;
3. Removing a bond.

The agent optimizes a multi-objective reward function designed to maximize:

- **Prediction Change**: Maximizing the difference between the original DGN prediction and the counterfactual one;

- **Similarity Preservation**: Minimizing structural differences by using a convex combination of Tanimoto similarity and cosine similarity.

In brief, the agent's goal is to alter the molecule **as little as possible**, while ensuring the modification is sufficient to flip its toxicity classification.

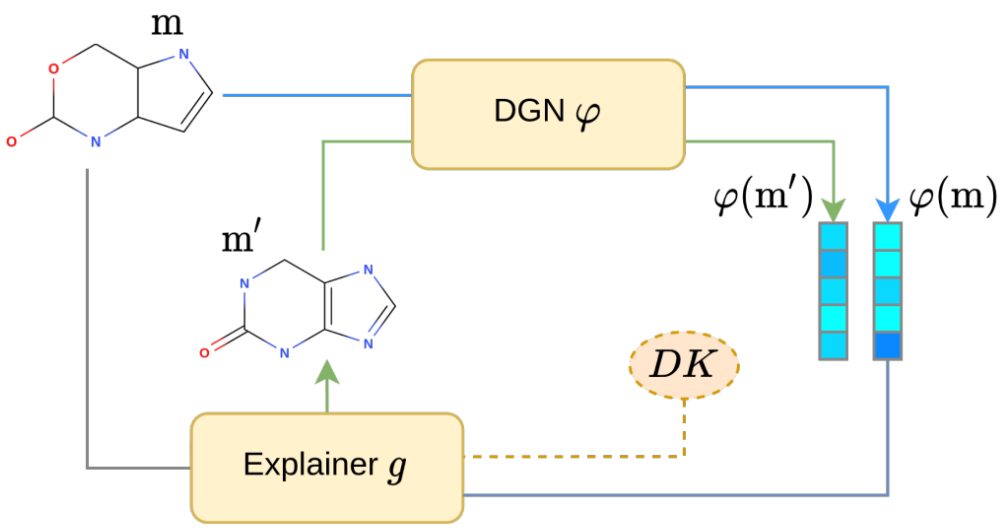

Figure 1: Model Architecture

Final Aims:

- Comparing the original and counterfactual molecular graphs to identify the specific atoms and bonds responsible for transforming a non-toxic molecule into a toxic one.
- Analyzing the model's decision boundary by observing the effects of the **discrete perturbations** applied by the generator agent.

Extensions: Possible extensions of this project include applying the DGN to regression tasks, such as predicting drug solubility, or studying Adverse Drug Reactions (ADR).

References

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